07A — Deep-Dive Feature Audit: Sect Governance & Internal Ecosystem (Part 1 of 5)

Sect Leadership, 6 Directorates & 25 Official Posts (`POST_CATALOG`), Autonomous Jurisdictions, Stipend Wallets, CA/PA Applicant Appraisal & Anti-Corruption Audits

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Target Audience**: Autonomous AI Developers (**GPT** & **Gemini**)
- **Core Game Philosophy**: **100% Emergent, Autonomous Xianxia Sandbox**. Every bureau unlock, appointment, daily jurisdiction, stipend payment, corruption scheme, disciple petition, and censor audit must emerge naturally from NPC psychology (`Human Core`), skills, relationships, and closed-loop sect finances—never locked behind player UI buttons or leaking Spirit Stones into dead fields!
- **Master Roadmap for Category 07 (`07A–07E`)**:
  - **`07A` (This Document)**: **Leadership, 6 Directorates & 25 Official Posts (`POST_CATALOG`), Officer Jurisdictions, Stipends, Applicant CA/PA Appraisal & Censor Audits**
  - `07B`: The 8 Sacred Peaks (`SECT.PEAKS`), Peak Vaults, 43-Path Sub-Sanctuaries, Peak Arrays, Expeditions & 144 Peak Stations
  - `07C`: Sect Buildings, Dormitory Housing Caps, Feng Shui 10-Element Matrix, Leyline Grid, Elemental Barrier, Ancestral Hall & Shrines
  - `07D`: Internal Politics, 4 Elder Cliques, Standing Edicts, 5 Sect Charters, Strategic Council, Tribunal, Dungeon Prisoners, Succession & Mutiny
  - `07E`: Grand Finale — Holistic Overview of the Entire Living Sect Ecosystem (Macro Fiscal Balance, Full NPC Lifecycle & Master `P0->P3` Roadmap)

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0. Hard Architectural Invariants

1. **NEVER modify `js/human_core.js`** (`HC.FINGERPRINT === '155:2f9c9dd7'`).
2. **Strict CA/PA Privacy Boundary**: `SECT.observableCapability` must **never** read hidden `aptitudes.*`, `affinities.*`, `intellect.*`, or `destiny`.
3. **Closed-Loop Fiscal Conservation**: Officer duties, stipends, and anti-corruption confiscations must conserve Spirit Stones (`Debit === Credit`) across `W.sect.stones`, **`npc.inv.stones`**, and `W.world.resourcePool.stones`.

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1. [GOV-OFF-01] Specialized Bureau Lockout: 9 of the 25 Offices Remain Disabled Unless Unlocked Autonomously as the Sect Grows!

- **Locations**: `js/sect.js` (`SECT.POST_CATALOG`, `SECT.officeState`, `SECT.unlockAllOffices`), `test/v310_autonomous_governance_and_peaks_sanctuary_suite.js:54–59`
- **Root Cause & Chain-Reaction Impact**:
  - `v3.1.0` added `9` specialized offices (`envoy`, `spymaster`, `beastmaster`, `arraymaster`, `censor`, `quartermaster`, `vanguard`, `warden`, `marketmaster`) gated behind `W.sect.unlockSpecializedOffices` / `officeState(W, id).enabled` (triggered via UI button 🔓 ปลดล็อกทำเนียบตำแหน่งทั้งหมด).
  - In a hands-off natural playthrough where the player never clicks that button:
    1. If `SECT.daily` / `SECT.masterDecision` does not progressively unlock specialized bureaus as population (`>= 14..28`), rank-3+ elders, and sect fame (`>= 60..200`) cross organic thresholds, **`36%` (`9 / 25`) of all sect offices stay permanently vacant**!
    2. Worse: **`SECT.STRATEGIC_COUNCIL` (`v4.0.0`)** explicitly requires the **6 Grand Officers (`spymaster`, `envoy`, `quartermaster`, `vanguard`, `warden`, `marketmaster`)**—all 6 of whom belong to the locked specialized tier!
- **Exact Fix**: Keep `SECT.unlockAllOffices(W)` for tests/UI, **and** add organic milestone auto-unlocking inside `SECT.daily(W)` (`e.g. warden` unlocks automatically the moment the first dungeon prisoner is captured or `pop >= 14`; `vanguard` at `pop >= 15` or first raid; `marketmaster` & `quartermaster` at `pop >= 18`; `envoy`, `spymaster`, `censor`, `arraymaster`, `beastmaster` as matching buildings/elders appear!).

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2. [GOV-STIPEND-02] Critical Split-Wallet Black Hole: `SECT.payOfficeStipends` Credits Dead `officer.personalStones` Instead of `officer.inv.stones`!

- **Locations**: `js/sect.js` (`SECT.payOfficeStipends`), `test/v310_autonomous_governance_and_peaks_sanctuary_suite.js:124–131` vs `js/external_market.js:405`, `js/cpeco.js:304`, `js/ai.js:893`, `test/v311:100`

- **Repository-Wide Proof**:
  - Full-codebase search for `personalStones` matches **only** `sect.js`, `visitors_clues.js`, and `v310`!
  - Every single gameplay consumer of an NPC's personal Spirit Stones—Visiting Merchant purchases (`external_market.js:405`), 9-Realm Specialization Track gates (`CULT.checkTrackGates`), CP stone grants (`cpeco.js:304`), Peak distributions (`v311:71`), monthly base salaries (`v311:100`), and death inheritance (`ai.js:893`)—reads and writes **`npc.inv.stones`**!
  - **Impact**: When `SECT.payOfficeStipends(W)` deducts **`~680` Spirit Stones/month** from `W.sect.stones` and writes them to `officer.personalStones` without crediting `officer.inv.stones` (or synchronizing the two), **hundreds of Spirit Stones every month disappear from the Sect Treasury into an unspendable, non-inheritable black hole**!
- **Exact Fix**:
  1. Credit `officer.inv.stones = (officer.inv.stones || 0) + bonus` inside `SECT.payOfficeStipends`, and define `officer.personalStones` as a synchronized mirror (`officer.personalStones = officer.inv.stones`) so `v310:131` (`pool[0].personalStones > cmdPersonalBefore`) continues passing 100%!
  2. Integrate `SECT.payOfficeStipends(W)` cleanly into `SIM.monthly` (`when W.sect.stones > cashReserveFloor`) alongside `SECT.paySalaries(W)` so officers are compensated organically every month without requiring manual UI clicks!

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3. `[GOV-JURIS-03]` Closed-Loop Spirit-Stone Leaks Inside Officer Jurisdictions & Censor Confiscations (`sect.js` & `v311:150–153`)!

2. **Quartermaster & Marketmaster Daily Stone Minting (`v311:150–153`)**:
   - `v311` lines `150–153` show `SECT.runOfficerJurisdictionSingle(W4, 'quartermaster', qmHolder)` increasing `W4.sect.stones` on a fresh Day-0 world with zero trade goods sold and zero external counterparty!
   - **Rule**: Whenever `quartermaster` (`logistics tariff / supply optimization`) or `marketmaster` (`surplus trade`) generates Spirit Stones for `W.sect.stones`, **it MUST debit `W.world.resourcePool.stones`** (or an active visiting merchant's `v.stones` in exchange for real surplus stock above `EXT.reserveForSect`), **never** minting free stones out of thin air!
   - Also verify `marketmaster` gear liquidation obeys the exact same protections as `EXT.sellStaleSectGear` (`skip playerSpawned, legendary, weaponSpirit, imbuedElderSoul, wantedUids, and ageDays < 6`)!

2. **Censor Audit Free-Stone Conjuration (`SECT.ELDER_COUNCIL_EDICTS.auditOfficerConduct`, `v2.0.32`)**:
    - In `auditOfficerConduct`:
        - `EXEMPLARY_STEWARDSHIP` (`integrity >= 68`) adds `+35` stones to `W.sect.stones` out of thin air! Instead, record an upkeep rebate credit (or debit `W.world.resourcePool.stones` as external supplier rebates).
        - `CLIQUE_NEPOTISM_DETECTED` (`integrity < 44`) adds `+45` "confiscated" stones to `W.sect.stones` **without checking or deducting `offender.inv.stones`**! If a broke officer has `0` `inv.stones`, catching them embezzling **prints `+45` free Spirit Stones out of thin air**! Confiscate `const seized = Math.min(45, offender.inv?.stones || 0)` from `offender.inv.stones` into `W.sect.stones` (plus file a real case in `SECT.TRIBUNAL.fileCase`!).

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4. `[GOV-HALL-04] 9` of the `25` Offices Reference Nonexistent Building IDs in `POST_CATALOG[id].hall`!

Comparing `SECT.POST_CATALOG` (`PATCH_IMPLEMENTATION_REPORT_V3_1_0.md` Table 2.1) against actual sect map building IDs (`W.map.byId` in `world_map.js` / `economy.js` / `ai.js`):

Office ID	Declared `p.hall`	Actual `W.map.byId` Building ID	Runtime Consequence
`commander`	`'command'`	`'main_hall'` / `'training_ground'`	`W.map.byId['command']` is `undefined`!
`martial`	`'martial'`	`'training_ground'`	`W.map.byId['martial']` is `undefined`!
`envoy`	`'diplomacy_hall'`	`'main_hall'` / `'tea'`	`W.map.byId['diplomacy_hall']` is `undefined`!
`spymaster`	`'intel_hall'`	`'main_hall'` / `'ruin'`	`W.map.byId['intel_hall']` is `undefined`!

Office ID	Declared `p.hall`	Actual `W.map.byId` Building ID	Runtime Consequence
`mission`	`'mission'`	`'main_hall'` / `'gate'`	`W.map.byId['mission']` is `undefined`!
`arraymaster`	`'array_pagoda'`	`'forge'` / `'shrine'`	`W.map.byId['array_pagoda']` is `undefined`!
`herbal`	`'herb'`	**`'herb_garden'`**	Typo! Misses the actual Herb Garden!
`beastmaster`	`'beast_park'`	**`'beast_menagerie'`**	Typo! Misses the actual Beast Menagerie!
`steward`	`'workshop'`	**`'mech_workshop'`** / **`'main_hall'`**	Misses actual workshop/hall!

- **Fix**: Add a `SECT.resolveOfficeHall(W, postId)` resolver mapping each office's logical `p.hall` to the real active `W.map.byId` building so `A.duty.where` physically stations the Beastmaster at `beast_menagerie`, the Herbal Elder at `herb_garden`, the Martial Elder at `training_ground`, and Vanguard/Guard/Wildwarden at `gate`!

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5. `[GOV-FIT-05]` Broken Skill Keys, 9-Weapon Blindness & 31-Path Suitability Gaps + Orphaned `W.sect.petitions` Queue!

1. **Wrong Skill Names & Missing 9 Weapons (`js/sect.js:76, 260`)**:
   - `SECT.observableCapability` checks `sk.LIGHTFOOT`, `sk.HERB_LORE`, and `sk.farming` (**none of which exist**—real keys are `MOVEMENT` and `herbalism`!).
   - Both `observableCapability` and `SECT.relevance` / `POST_CATALOG.suitability` check only the 7 legacy weapon keys (`SWORD..FOCUS`), giving **`0` martial capability** to masters of `ZITHER`, `BRUSH`, `WHIP`, `ROSARY`, `CLAW`, `NEEDLE`, `FAN`, `ASTROLABE`, and `ABACUS` when evaluating `commander`, `vanguard`, `martial`, `guard`, and `discipline`! Use `MARTIAL.maxWeaponSkill(n)` across all 16 weapon families (while strictly

keeping `observableCapability` free of forbidden `aptitudes`/`affinities`/`intellect`!).

2. **Integrate All 31 Grand Dao Paths Into `POST_CATALOG[id].suitability`**:
   - Grant thematic competence bonuses (`+12..+20` suitability) for Grand Dao specialists: `MERCANTILE_DAO` $\to$ `marketmaster`/`treasury`; `NINPO_SHADOW` $\to$ `spymaster`; `LORE_ARCHIVIST` & `INK_LITERATI` $\to$ `scripture`/`registry`/`censor`; `DRAGON_VEIN_GEOMANCY` & `ASTRO_DIVINATION` $\to$ `arraymaster`; `SPIRIT_HUSBANDRY` $\to$ `herbal`/`provision`; `BUDDHA_DHARMA` $\to$ `healer`/`discipline`.
3. **Connect the Broken `W.sect.petitions` Causal Chain (`cpeco.js:495` $\to$ `cognition.js:1005` $\to$ `censor` / `discipline`)**:
   - Currently, injured disciples passed over by a nepotistic `treasury` officer file formal complaints into `W.sect.petitions`, **which nobody ever reads (`02B`)**!
   - **Fix**: Inside `SECT.runOfficerJurisdictionSingle` for **`censor`** (and **`discipline`**), dequeue pending complaints from `W.sect.petitions`! If the `censor` is righteous (`integrity >= 58`), they escalate corroborated `unfair_allocation` petitions into **`SECT.TRIBUNAL.fileCase(W, accusedOfficer, 'FACTION_CORRUPTION')`**; if the `censor` is corrupt (`integrity < 42`) and in the **same political clique** as the accused treasurer, they **cover up the petition**, lowering complainant `loyalty -6` and fueling `FACTION_MUTINY` undercurrent!

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6. Mandatory Verification Suite Additions (`07A`)

1. **Stipend Wallet Sync Test**: Run `SECT.payOfficeStipends(W, { forceFull: true })` and assert **both** `officer.inv.stones` and `officer.personalStones` increase by the exact amount deducted from `W.sect.stones`.
2. **Closed-Loop Jurisdiction & Audit Test**: Run all 25 `runOfficerJurisdictionSingle` duties and `auditOfficerConduct` on a world; assert `W.sect.stones + sum(npc.inv.stones) + W.world.resourcePool.stones + sum(visitor.stones) + sum(peak.treasuryStones)` is **100% conserved to the single stone**!
3. **Petition-to-Tribunal Causal Loop Test**: Trigger an `unfair_allocation` petition in `W.sect.petitions`, run `censor` jurisdiction with high `integrity`, and assert a `FACTION_CORRUPTION` case is opened in `W.sect.tribunal.activeCases`.